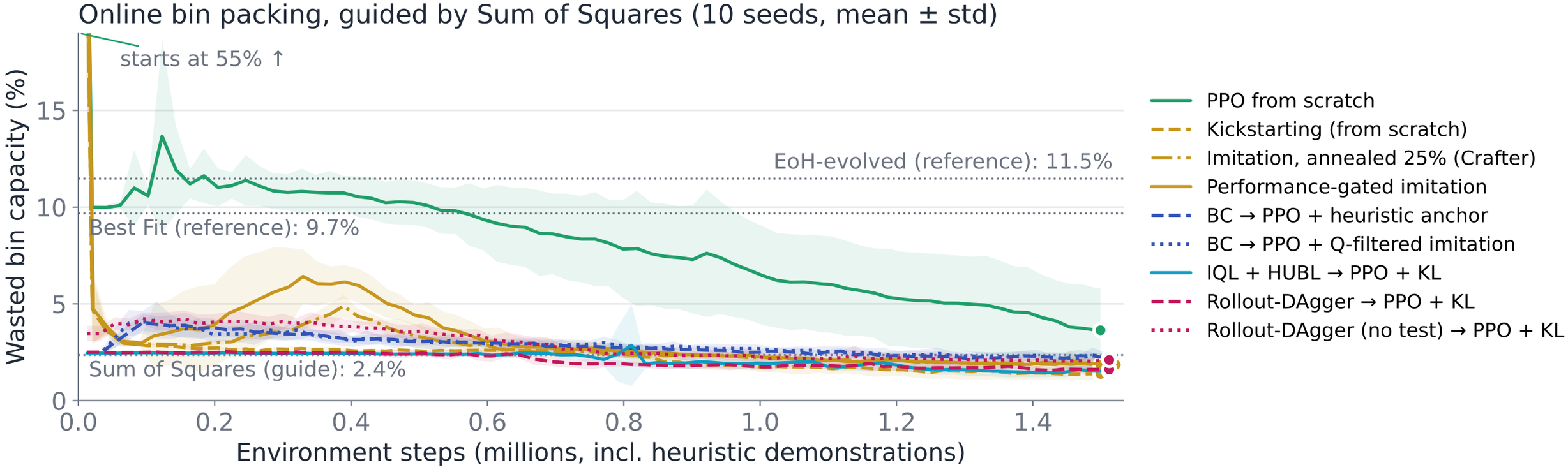
Online bin packing, guided by Sum of Squares (10 seeds, mean ± std)
starts at 55% ↑
EoH-evolved (reference): 11.5%
Best Fit (reference): 9.7%
Sum of Squares (guide): 2.4%
Wasted bin capacity (%)
0
5
10
15
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
Environment steps (millions, incl. heuristic demonstrations)
PPO from scratch
Kickstarting (from scratch)
Imitation, annealed 25% (Crafter)
Performance-gated imitation
BC → PPO + heuristic anchor
BC → PPO + Q-filtered imitation
IQL + HUBL → PPO + KL
Rollout-DAgger → PPO + KL
Rollout-DAgger (no test) → PPO + KL